

AIT352 Artificial Neural Networks Techniques

Question Bank

Module-I

1. Differentiate between biological neuron and artificial neuron?
2. Draw a simple artificial neuron and discuss the calculation of net input?
3. Differentiate between supervised and unsupervised learning?
4. What is an activation function? What is its necessity?
5. Explain various types of connections in artificial neural networks?
6. State the training algorithm used for the Hebb network?
7. Explain the architecture of McCulloch-Pitts neuron?
8. Briefly explain about the characteristics of artificial neural networks.
9. Advantages of neural network
10. Explain about the application scope of artificial neural networks
11. Implement AND function using McCulloch-pitts neuron (take binary data)
12. Define i) weight ii) bias iii) threshold iv) Learning rate
13. Architecture of MP neuron
14. Concept of linear separability
15. Obtain the output of the neuron for a network with inputs are given as $[x_1, x_2] = [0.7, 0.8]$ and the weights are $[w_1, w_2] = [0.2, 0.3]$ with bias = 0.9. Use i) Binary sigmoidal activation function ii) Bipolar sigmoid activation function
16. What are the different types of competitive learning?
17. Demonstrate the significance of different Activation functions.
18. Explain the terms cell body, axon, synapse, dendrite and neuron with reference to biological neural network.
19. What are the main differences among the three models of artificial neurons, namely, McCulloch-Pitts, Perceptron and Adaline?
20. Compare the stability and convergence of ANN
21. Compare the performance of a computer and that of a biological neural network in terms of speed of processing, size and complexity, storage, fault tolerance and control mechanism.
22. What is reinforcement learning? In what way it is different from supervised learning?

23. Describe any four attractive features of the biological neural network that make it superior to the most sophisticated Artificial Intelligence computer system for pattern recognition tasks

23. Explain about different learning tasks

24. Briefly explain knowledge representation

25. Differentiate Hebbian learning and Competitive learning

Module-2

1. Explain about perceptron network with figure (architecture)

2. Explain about perceptron learning rule

3. Explain perceptron training and testing algorithm for single output classes with flowchart

5. Explain Adaline architecture.

6. Briefly explain on delta rule/ Widrow Hoff rule in Adaline

7. Explain training and testing algorithm of Adaline network with flowchart.

8. Design a McCulloch Pitts neural network to implement AND function.

9. Define Perceptron Learning Rule Convergence theorem

10. Discuss about Adaptive linear neuron (Adaline)?

11. State the testing algorithm used in perceptron network?

12. Explain the training algorithm used in Adaline network?

13. Explain the perceptron training algorithm for single output classes?

14. State the testing algorithm for Adaptive Linear Neuron (Adaline)?

15. Explain delta rule?

16. Mention the applications of perceptron network.

17. Implement AND function using perceptron networks for bipolar inputs and bipolar targets (assume $\alpha=1$)

18. Using the Perceptron Learning rule find the weights required to perform the following classifications. Vectors (1,1,1,1) and (-1,1,-1,-1) are members of the class and hence target value 1; vectors (1,1,1,-1) and (1, -1, -1, 1) are not the members of the class and hence target value -1. Use learning rate of 1 and starting value of weights as 0, test the response of the net.

19. Find the weights using perceptron network for ANDNOT function when all the inputs are presented only one time. Use bipolar inputs and targets.

20. Implement OR function with bipolar inputs and targets using Adaline network. (Use two epochs- assume $w_1=w_2=b=0.1$ and $\alpha=0.1$)

21. Use Adaline network to train ANDNOT function with bipolar inputs and targets. Perform two epochs. (Assume $w_1=w_2=b=\alpha=0.2$)

22. XOR function is non linearly separable by a single decision boundary line. Justify

23. Implement OR function with binary inputs and bipolar targets using perceptron training algorithm upto 3 epochs (Assume $\alpha=1$ and $\theta=0.2$)

24. How is the linear separability concept implemented using Perceptron Network training

25. Using Hebb Network calculate the weight required to perform the following classification of given input pattern. The inputs are “1” where “+” symbol is present and “-1” where “*” is present

L belongs to the members of the class(+), target value +1

U does not belong to members of class(*), target value -1

+	*	*
+	*	*
+	+	+

L

+	*	+
+	*	+
+	+	+

U